

Executive Business Report

Retail Sales Forecasting and Demand Analysis

Executive Summary

This report presents the results of a retail sales forecasting and demand analysis project aimed at improving inventory planning and business decision-making. Historical sales data was analyzed to identify sales trends, forecast future demand, detect unusual sales events, and segment products based on demand patterns. Among the forecasting models evaluated, **Prophet** achieved the best performance and was selected for deployment. The analysis indicates that the **Technology** category and the **West** region are expected to experience the strongest growth over the next three months. These insights can help the business optimize inventory levels and improve supply chain planning.

Key Findings from EDA and Forecasting

The exploratory analysis showed that the **Technology** category generated the highest total revenue, followed by **Furniture** and **Office Supplies**. Region-wise analysis revealed that the **West** region consistently recorded the highest sales.

Time series analysis identified an overall upward sales trend with seasonal variations. Three forecasting models were compared:

Model	MAE	RMSE	MAPE
SARIMA	15,062.03	18,963.90	19.59%
Prophet	14,309.99	18,954.58	17.47%
XGBoost	19,150.47	21,282.98	22.96%

Based on these results, the **Prophet** model was selected because it produced the most accurate forecasts.

Three-Month Sales Forecast

The three-month forecast indicates continued business growth, with the **Technology** category expected to achieve the strongest increase in sales. The **West** region is also forecasted to remain the highest-performing geographical market. Forecast confidence intervals provide a realistic range of expected sales, allowing management to plan inventory while accounting for forecasting uncertainty.

Top Three Anomalies Detected and their likely cause

Two anomaly detection techniques (Isolation Forest and Z-Score) identified unusual weekly sales patterns.

- March 2015:** A significant sales spike, likely due to promotional campaigns or bulk purchases.
- October–November 2018:** Multiple high-sales weeks, likely associated with festive or holiday shopping.
- Several low-sales weeks:** Possibly caused by inventory shortages, shipment delays, or temporary operational disruptions.

Isolation Forest detected both unusually high and unusually low sales periods, while the Z-Score method mainly identified high-sales spikes.

Product Demand Segmentation and recommended stocking strategy per segment

K-Means clustering grouped product sub-categories into three demand segments:

- **Premium / High Value Products:** Maintain safety stock to prevent revenue loss from stock-outs.
- **Stable Everyday Demand:** Use regular replenishment and standard inventory levels.
- **High Volume & Growing Demand:** Maintain higher inventory levels and prioritize replenishment.

This segmentation supports more effective inventory management and demand planning.

3 concrete business recommendations with data backing each one

1. **Increase inventory for Technology products**, as they generate the highest revenue and are forecasted to experience the strongest future growth.
 2. **Prioritize inventory allocation in the West region**, where historical and forecasted sales remain consistently high.
 3. **Use forecasting and anomaly monitoring together** to improve inventory planning, identify unusual sales events early, and reduce stock shortages.
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Risk / Limitation of this system that the business should be aware of

Forecast accuracy depends on historical sales patterns. External factors such as economic conditions, promotional campaigns, supplier disruptions, and changing customer preferences may affect future sales. Regular model updates with new data are recommended.

Conclusion

The project demonstrates that data-driven forecasting and demand analysis can improve inventory planning and business decision-making. The Prophet model delivered the best forecasting performance, while anomaly detection and product demand segmentation provided additional insights for supply chain optimization. Implementing these findings can help the organization improve stock availability, reduce operational risks, and support long-term business growth.
